

SANIYA MARIA SUNIL

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PROFESSIONAL SUMMARY

AI and Data Engineering student at Nanyang Polytechnic. Experienced in building end-to-end IoT and data systems, from wearable sensor collection over BLE to SQLite storage and a Flask dashboard, and deployed a cloud application on AWS, using Python, MySQL, Arduino and Raspberry Pi. Reliable and detail-oriented, with a year of customer-facing work handling cash, records and safety procedures, plus peer tutoring and volunteering experience.

EDUCATION

Nanyang Polytechnic 2025-2028
Diploma in AI and Data Engineering

CHIJ St Joseph's Convent 2021-2024
O-Level

TECHNICAL SKILLS

- **Programming & Data:** Python, MySQL, SQLite, Microsoft Excel, Google Colab
- **IoT & Hardware:** Arduino, Raspberry Pi, BLE wearable sensors (accelerometer and gyroscope), ultrasonic and gesture sensors
- **Cloud & Web:** AWS (web server, VPC, database, object storage), Flask, Render deployment
- **Design & Fabrication:** Autodesk Inventor, Bambu Lab 3D printing
- **Coursework:** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing

PROJECTS

Construction Site Safety Monitoring, IoT Data Engineering Project

- Built a wearable system using four Nesso devices (BMI270 accelerometer and gyroscope, 2 Hz) that streamed worker movement data over Bluetooth Low Energy to a Python collector
- Validated and classified readings with threshold-based rules into Fall From Height, Slip-Trip-Fall and Near Miss events, stored in SQLite
- Developed a Flask dashboard to search by worker, filter by date, time and event type, and analyse incident trends; deployed on Render for remote historical review
- Monitored 4 workers in testing and recorded 1,135 Near Miss, 700 Slip-Trip-Fall and 190 Fall From Height events

Cloud Computing Project, E-commerce Application on AWS

- Designed and deployed a secure, cost-effective cloud application with a web server, VPC, database and object storage
- Displayed nine products on the home page, with product data in the database and images in object storage
- Completed within the $50 budget and project timeframe

Touchless Smart Lighting System, IoT Prototype Development

- Built an automatic lighting prototype: an ultrasonic sensor switches the LED on and off by presence, and a gesture sensor adjusts brightness without touching a switch

Other projects: Python fast-food ordering and payment system with discounts and GST, original 3D model designed in Autodesk Inventor and printed on a Bambu Lab printer

IoT & Hardware: Arduino, Raspberry Pi, BLE wearable sensors (accelerometer and gyroscope), ultrasonic and gesture sensors
Cloud & Web: AWS (web server, VPC, database, object storage), Flask, Render deployment
Design & Fabrication: Autodesk Inventor, Bambu Lab 3D printing
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WORK EXPERIENCE

Kids Club / Park Operator, Nestopia, Shangri-La Rasa Sentosa **Feb 2025 – Present**

- Handled cashier duties and answered booking enquiries by email with accurate information on park services
- Prepared and maintained Kids Club and park documents and records using Microsoft Excel and Word
- Ran counter closing, including cash reconciliation and end-of-day reporting, with accuracy and accountability
- Supervised children during play sessions and enforced safety procedures, keeping the experience positive for tourists, hotel guests and parents

AWARDS AND CERTIFICATIONS

Edusave Certificate of Academic Achievement 2026
Basic Proficiency in KNIME Analytics Platform (L1) | KNIME AG 2026
GenAI Powered Data Analytics Job Simulation | Tata Group via Forage 2026
AI Fluency Framework & Foundations Anthropic 2025

CO-CURRICULAR ACTIVITIES

Mentorship Club Subcommittee Member 2026

- Volunteer with KISREAD, supporting Primary 1–6 students in developing their English reading and writing skills.
- Participate in SGSS Mentoring, providing academic guidance and support to students in subjects they find challenging.
- Organise and facilitate Level-Up events, encouraging NYP students to participate in activities that develop new skills and enhance personal growth.

School Of Engineering Club (SEG) Subcommittee Member 2026

- Volunteered for NYP open house promoting the school , and my course individually
- Student Mentor for Freshmen Orientation 2026
- Volunteered for Community Learning Fest for children to learn more about 3D design and modelling

PRIMERS Volunteer 2025

- Collected and delivered food donations to low-income households, supporting families in need through community outreach initiatives.
- National Day Parade (NDP) Motivator. Served as a motivator during the National Day Parade through PRIMERS, encouraging and engaging participants to contribute to a lively and meaningful celebration.

IJHCC – VIA Project Volunteer 2024

- Planned and conducted a Games Day for children at IJHCC as part of a Values-in-Action (VIA) project in 2024.
- Facilitated games and interactive activities to create an enjoyable and engaging experience for the children.

Willing Hearts – Volunteer April 2026- Present

- Volunteer at Willing Hearts whenever available, contributing time and effort to support its community food distribution efforts.